

FAST National University of Computer and Emerging Sciences

Karachi Campus

Mental Health Support App Prototype for Students Using Figma

Findings and Discussion

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1 Findings and Discussion

This section presents the results obtained from a survey conducted among university students at FAST National University of Computer and Emerging Sciences (NUCES), Karachi Campus. A total of 19 responses were collected through a structured questionnaire distributed via Google Forms. The findings are organized according to the key themes of the survey and are discussed in relation to the existing literature on student mental health and digital application design.

1.1 Demographic Overview

The majority of respondents (84.2%) were enrolled in the third or fourth year of their degree programs, with the remaining 15.8% being first or second year students. In terms of gender, 13 respondents identified as male and 6 as female. All participants were from the Engineering and Computer Science discipline, with one exception from Business and Management. This sample, while limited in disciplinary diversity, provides a focused insight into the mental health experiences of students in a highly demanding academic program.

1.2 Prevalence of Stress and Anxiety

The survey results reveal a widespread experience of stress and anxiety among university students. Only one respondent (5.3%) reported never experiencing stress. The remaining 94.7% reported experiencing stress at varying levels: 15.8% rarely, 36.8% sometimes, 21.1% often, and 21.1% always. These figures are consistent with existing literature, which identifies university students as a population at elevated risk for psychological distress due to academic and social pressures [1].

Career and future uncertainty emerged as the most frequently cited source of stress, reported by 16 out of 19 respondents (84.2%). Academic workload and deadlines followed closely, cited by 13 respondents (68.4%). Family expectations were identified by 7 respondents (36.8%), financial pressure by 5 (26.3%), and social relationships by 2 (10.5%). These findings suggest that the primary stressors for students in this context are external and future-oriented rather than immediately interpersonal, which has implications for the types of support features that should be prioritized in a mental health application.

1.3 Help-Seeking Behaviour

A significant finding of this study is the low rate of professional help-seeking among the respondent group. Only 3 out of 19 respondents (15.8%) reported having sought

professional mental health support such as counselling or therapy. Four respondents (21.1%) stated that they had wanted to seek help but did not, while the majority (63.1%) had never sought professional support at all.

When asked about the reasons for not seeking help, the most dominant response was a preference for self-management, cited by 11 respondents (57.9%). Four respondents indicated they did not know where to go for support, three cited financial barriers, and one cited fear of social judgment or stigma. These findings align with prior research indicating that stigma, lack of awareness, and structural barriers are primary obstacles to mental health help-seeking among university students [2]. Importantly, the high proportion of students who prefer to manage their mental health independently suggests a strong potential for self-help digital tools that do not require clinical referral or in-person interaction.

1.4 Current Digital Tool Usage

Despite the high prevalence of stress reported, 89.5% of respondents indicated that they do not currently use any mobile application to manage their mental health or well-being. Only 2 respondents reported using an app, and both rated their experience as neutral or very dissatisfied. This near-total absence of current app usage, despite significant unmet mental health needs, highlights a critical gap in accessible and engaging digital mental health support for this student population.

Respondents spent considerable time on their smartphones, with the majority (52.6%) reporting more than 6 hours of daily smartphone usage, and a further 26.3% reporting between 4 and 6 hours. This level of smartphone engagement confirms that a mobile application is an appropriate and accessible delivery format for mental health support, consistent with findings in the mHealth literature [3].

1.5 Feature Preferences

Respondents were asked to rank six proposed application features from 1 (most useful) to 6 (least useful). The average rankings were computed to determine relative feature priority. The results are presented in Table 1.

Table 1: Average Feature Rankings (1 = Most Useful, 6 = Least Useful)

Feature	Average Rank
Progress Tracker & Reminders	2.89
Journaling / Daily Diary	3.00
Mood Tracking	3.21
Emergency Contact / Crisis Support	3.21
Mental Health Resources & Articles	3.47
Guided Relaxation & Breathing Exercises	3.53

Progress Tracker and Reminders received the highest average priority ranking, followed by Journaling and Mood Tracking. These results suggest that students value tools that help them monitor their emotional patterns and maintain consistency over time. Guided Relaxation and Mental Health Resources were ranked comparatively lower, though this does not indicate low importance, but rather that students consider tracking and journaling features more immediately practical in their daily routines.

All respondents (100%) stated that ease of use and visual simplicity were either important or extremely important. This is a critical design requirement and reinforces the HCI literature’s emphasis on simplicity and reduced cognitive load in mental health application design [4].

1.6 Anonymity and Usage Preferences

When asked about their preferred mode of interaction, 8 respondents (42.1%) preferred a fully anonymous experience with no account required, 7 (36.8%) preferred a personalized profile with login, and 4 (21.1%) indicated that either option would work. The near-even split between anonymity and personalization suggests that an ideal design would offer both modes, allowing users to choose their preferred level of privacy. This finding is particularly significant in the Pakistani cultural context, where stigma around mental health remains a notable barrier to help-seeking.

Regarding anticipated usage frequency, 4 respondents indicated they would use the app once a day consistently, 4 said a few times a week, and the remaining majority indicated rare or occasional usage. This suggests that features such as reminders and push notifications could play a key role in improving retention and consistency of use.

1.7 Qualitative Insights

The open-ended responses provided rich qualitative data that complement the quantitative findings. Several recurring themes emerged from responses regarding desired features and frustrations with existing applications.

A prominent theme was the desire for **anonymity and non-judgmental interaction**. One respondent expressed a need to talk to someone anonymously, noting a lack of an outlet free from preconceived judgments. This reinforces the importance of designing a safe and stigma-free digital space for students.

A second theme was frustration with **generic and impractical content**. Multiple respondents criticized existing applications for offering generic advice with no room for personalization. One respondent specifically highlighted the frustration of re-explaining their situation each time they use an app, calling for context-aware tools that remember user patterns over time.

A third theme was the need for **cultural relevance**. One respondent suggested that the app include motivational quotes, Quranic verses, or Hadiths, reflecting the importance of religiously grounded content for users in a Muslim-majority context. Another respondent called for tools that fit student life and know when to escalate to real help. These insights indicate that a generic, Western-designed mental health application may not be fully appropriate for this user group, and that cultural customization is a meaningful design consideration.

A fourth theme was the rejection of **overly clinical or sympathetic framing**. One respondent noted that users do not want to feel special; they want to feel normal. This suggests that effective design should avoid medicalizing the user experience and instead normalize emotional difficulties as part of everyday student life.

1.8 Summary of Discussion

The findings of this study collectively indicate a significant and unmet need for a student-centered, culturally relevant, and privacy-respecting mental health application at FAST-NUCES Karachi. Students experience high levels of stress but overwhelmingly prefer to manage these challenges independently, making a self-help digital tool highly appropriate. The near-absence of current app usage despite widespread need points to a design gap rather than a lack of demand.

Key design priorities identified from the data include: full ease of use and visual simplicity, support for anonymous access, progress tracking and journaling as core features, avoidance of generic or overly clinical content, and incorporation of culturally relevant motivational content. These findings directly inform the design decisions made in the Figma prototype developed as part of this research project, ensuring that the prototype is grounded in real user needs rather than assumptions.

References

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